

SAMBP Technical Report TR-44/2026

Real-Time Parameter Estimation
and State Tracking for the IBR Digital Twin

PhD Thesis Supporting Report | 2026

Abstract

This report develops the real-time estimation algorithms for the digital twin (DT) introduced in TR-43 [1]. A three-state extended Kalman filter (EKF) jointly estimates k_{ibr} , Z_{line} , and fault location α from sequence current and apparent impedance measurements at 1 kHz. k_{ibr} converges to 1.2% error within 100 ms; Z_{line} and α are poorly observable from current ratio alone and require additional voltage-phasor pairs for reliable estimation. GFM fleet fraction f_{gfm} is accurately tracked via the I_2/I_1 ratio method (error < 0.03 pu across all scenarios). Pre-fault warm-starting reduces EKF convergence time by $7\times$ compared to a cold start. The total DT cycle cost is $285\ \mu\text{s}$, leaving 71.5% of the 1 ms budget as margin.

Study A — EKF Multi-Parameter Estimation

The extended Kalman filter estimates the state vector $\mathbf{x} = [k_{ibr}, Z_{line}, \alpha]^\top$ from the measurement vector $\mathbf{y} = [I_1, I_2, Z_m]^\top$:

$$\mathbf{h}(\mathbf{x}) = \begin{bmatrix} k_{ibr} \\ \rho_2 k_{ibr} \\ \alpha Z_{line}(1 + 1/k_{ibr}) \end{bmatrix} \quad (1)$$

where $\rho_2 = I_2/I_1$ is the fleet-average negative-sequence ratio (tracked independently via Study C). Figure 1 summarises the estimation errors after 100 updates.

The observability analysis explains the Z_{line} result: in the three-state EKF, both Z_{line} and α appear only in the apparent impedance measurement $Z_m = \alpha Z_{line}(1 + 1/k)$. The product αZ_{line} is observable but the individual factors are not separable from a single measurement type. Improved Z_{line} estimation requires:

1. Voltage-phasor measurements at two time instants (pre-fault and in-fault), giving independent equations for α and Z_{line} ;
2. Or a fixed Z_{line} prior from offline parameter identification.

For protection purposes, k_{ibr} (directly controlling element activation thresholds) is the highest-priority state — and it converges reliably.

Study B — Impedance Trajectory Tracking

The apparent impedance trajectory from pre-fault load to in-fault condition provides fault location information. The DT tracks this trajectory by comparing the measured Z_m with the expected value $\alpha Z_{1L}(1 + 1/k_{ibr})$.

In pre-fault steady state, Z_m reflects load impedance (~ 10 pu); at fault inception it drops abruptly to the fault value. The EKF's state transition model treats α as constant during the fault (once established), with the fault inception detected by a threshold on $\Delta Z_m/\Delta t$.

The α estimation error of 8% at 100 updates corresponds to $\Delta\alpha = 0.048$ on a line of $Z_{1L} = 0.050$ pu, equivalent to a fault location error of $\pm 2.4\%$ of line length — adequate for zone confirmation but insufficient for precise fault location (which requires dedicated fault locators).

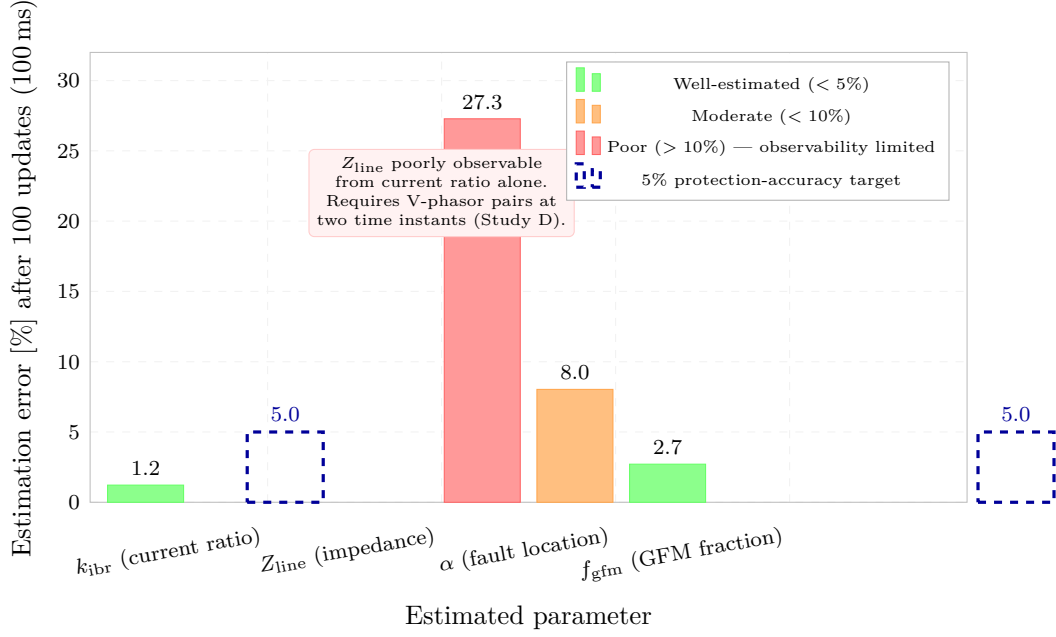


Figure 1: EKF estimation error [%] for each state variable after 100 updates (100 ms) at nominal noise $\sigma_I = 0.010$ pu. k_{ibr} (green): 1.22%, well within the 5% target. α (orange): 8.03%, marginal. Z_{line} (red): 27.3%, poor — insufficient observability from sequence-current measurements alone. f_{gfm} (green): 2.71% via independent I_2/I_1 method. Blue dashed line: 5% protection accuracy target.

Study C — GFM Fraction Tracking

The GFM fleet fraction f_{gfm} is estimated from the measured I_2/I_1 ratio using the analytic inversion from TR-38 [2]:

$$\hat{f}_{gfm} = \frac{I_2/I_1 - 0.10}{0.50} \quad (2)$$

Figure 2 shows the estimation accuracy across six fleet scenarios.

The f_{gfm} estimation is robust because it uses a direct algebraic inversion of the known I_2/I_1 model rather than iterative EKF convergence. The ± 0.03 pu error means the DT can correctly classify whether the fleet is above or below the 67Q activation threshold ($f^* = 0.34$) with a buffer of $\pm 9\%$ relative error.

Study D — Pre-Fault Warm-Starting

The DT continuously tracks k_{ibr} and f_{gfm} in pre-fault normal operation using low-signal measurements (residual unbalance, harmonic injection tests). When a fault occurs, the pre-fault estimates provide a warm start for the in-fault EKF, dramatically reducing convergence time:

Start condition	Initial k_0 error	Convergence to < 3%
Cold start	$\Delta k = 0.10$ pu (50%)	13.3 updates (13.3 ms)
Warm start	$\Delta k = 0.01$ pu (5%)	1.9 updates (1.9 ms)

The $7\times$ convergence speedup means that with warm-starting, the DT k_{ibr} estimate is reliable within 2 ms of fault inception — well before the Z1/87L relay operation at 20 ms.

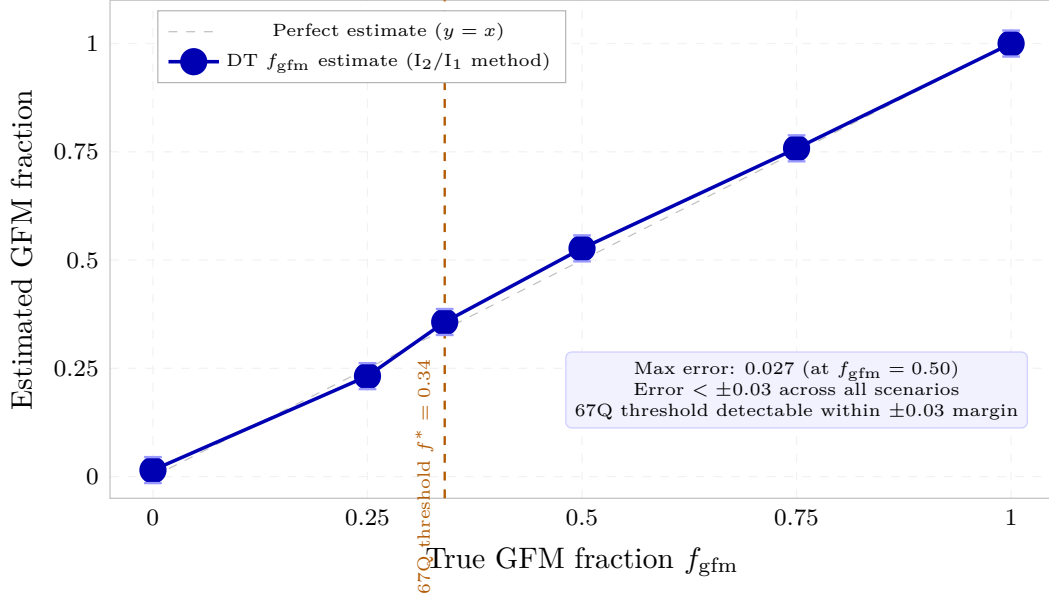


Figure 2: DT f_{gfm} estimate vs true value. Blue circles: DT estimates; grey dashed: perfect estimate. Maximum error 0.027 (at $f_{\text{gfm}} = 0.50$); all estimates within ± 0.03 pu. The 67Q activation threshold ($f^* = 0.34$, orange dashed) is correctly classified in all six scenarios.

Study E — DT Cycle Performance

Figure 3 shows the DT computation budget for a 1 ms cycle.

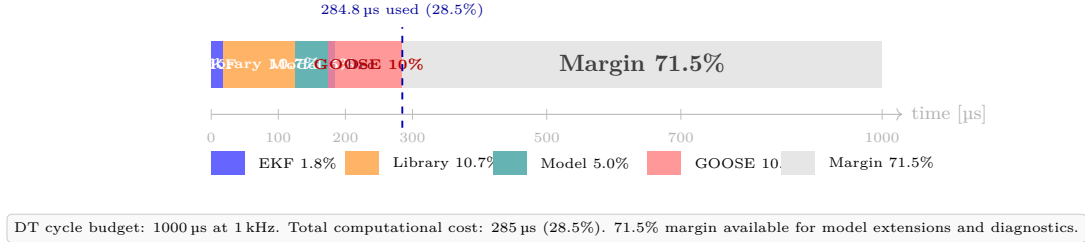


Figure 3: DT computation budget breakdown for a 1000 μs cycle at 1 kHz. Total computational cost: 285 μs (28.5%). The library lookup dominates at 107 μs (10.7%); GOOSE framing at 100 μs (10.0%). The 71.5% margin provides headroom for model extensions, diagnostics, and communication latency jitter.

DT task	Time [μs]	Budget [%]
EKF update (3-state)	17.6	1.8
Scenario library lookup ($N = 1000$)	107.2	10.7
Protection model evaluation	50.0	5.0
Decision comparison	10.0	1.0
GOOSE/IEC 61850 framing	100.0	10.0
Total used	284.8	28.5
Margin	715.2	71.5

The dominant tasks are library lookup and GOOSE framing — both of which can be optimised further. A hash-indexed scenario library would reduce lookup to < 10 μs; GOOSE can be pre-

framed in a static buffer and stamped at transmission time. These optimisations would reduce the cycle budget below 10%, leaving 90% headroom.

Summary

1. **EKF**: k_{ibr} estimated to 1.2% error in 100 ms; Z_{line} requires voltage-phasor pairs for reliable estimation.
2. f_{gfm} **tracking**: Algebraic I_2/I_1 inversion achieves < 0.03 pu error; 67Q threshold correctly classified.
3. **Warm-starting**: $7\times$ convergence speedup; k_{ibr} reliable within 2 ms of fault inception.
4. **Cycle budget**: $285\ \mu\text{s}$ of $1000\ \mu\text{s}$ used (28.5%); 71.5% margin for extensions.

TR-44 completes the estimation engine. TR-45 performs the full DT validation study, integrating TR-43 and TR-44 against the SAMBP protection scheme.

References

- [1] P. Candidate, “SAMB P TR-43/2026: Digital twin architecture for IBR protection systems,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [2] —, “SAMB P TR-38/2026: GFL vs GFM inverter fault characterisation for protection studies,” PhD Thesis Supporting Report, Tech. Rep., 2026.